Model Predictive Control (MPC) Design for an Inverted Pendulum System

1 Overview

This document details the design and implementation of a Model Predictive Control (MPC) strategy for an inverted pendulum system. The MPC framework leverages a discrete-time state-space model to predict future system behavior and optimize control actions over a finite horizon, while respecting input constraints and balancing performance with control effort.

2 Discrete-Time System Conversion

The continuous-time state-space model is converted to a discrete-time model using a sampling time $T_s = 0.1$ seconds:

$$\text{sys\_d} = \texttt{c2d(sys\_ss, Ts)}.$$

This step is essential since the MPC controller operates on a discrete-time basis.

3 MPC Controller Setup

An MPC controller object is created for the discrete model:

$$\text{mpc\_controller} = \texttt{mpc(sys\_d, Ts)}.$$

Key settings for the controller include:

- **Prediction Horizon:** $N_p = 20$ steps. This horizon determines how far into the future the controller predicts the system's behavior.
- **Control Horizon:** $N_c = 5$ steps. This defines the number of future control moves that are optimized at each step.

4 Weighting Parameters

Weighting parameters are used to tune the performance of the controller by penalizing deviations in outputs, excessive control actions, and rapid changes in control inputs:

- **Manipulated Variable Weight:** 0.1 Penalizes the magnitude of the control input.
- **Manipulated Variable Rate Weight:** 0.1 Penalizes rapid changes in the control input.
- **Output Variables Weights:** $[1 \quad 1]$ These weights ensure both the cart position and pendulum angle are regulated with equal importance.
- **ECR (Constraint Relaxation Weight):** 100000 A high penalty for constraint violation, ensuring the input constraints are strictly enforced.

5 Input Constraints

Physical limits are imposed on the control input (force applied to the wheels) to ensure practical operation:

$$-10 \leq F \leq 10.$$

These constraints are set within the MPC object to prevent unrealistic or damaging control actions.

6 Simulation Setup and Reference Trajectory

A simulation is conducted over a total time of $T_{\text{sim}} = 10$ seconds. The reference trajectory for the outputs is defined as:

$$r(t) = \begin{cases} 0, & \text{for } t < 1\,\text{second}, \\ 1, & \text{for } t \geq 1\,\text{second}. \end{cases}$$

This step input tests the controller's ability to track a sudden change in the desired outputs.

7 Closed-Loop Simulation and Analysis

The MPC controller is simulated using the defined parameters and reference trajectory. The simulation outputs include:

- **Closed-Loop Responses:** The time response of both the cart position and the pendulum angle.
- **Control Input:** The force applied by the MPC at each time step.

The resulting plots of the closed-loop responses and the control input provide insight into the controller's performance, ensuring that the system tracks the reference while obeying the input constraints.

8 Summary

The MPC controller design leverages a discrete-time model to predict and optimize future behavior over a finite horizon. By tuning the prediction and control horizons, weighting parameters, and incorporating physical input constraints, the controller is able to robustly track reference trajectories while mitigating excessive control actions. This balance of performance and robustness is key for practical implementation in systems such as the inverted pendulum.